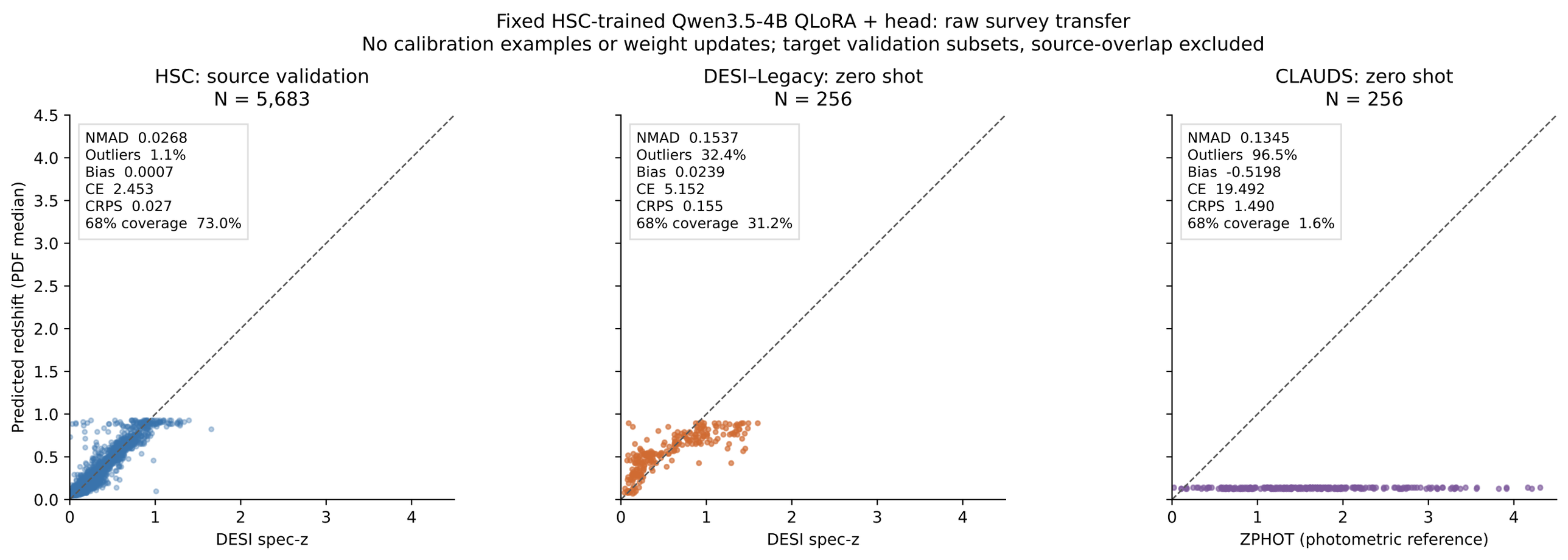
Fixed HSC-trained Qwen3.5-4B QLoRA + head: raw survey transfer
No calibration examples or weight updates; target validation subsets, source-overlap excluded
HSC: source validation
N = 5,683
NMAD 0.0268
Outliers 1.1%
Bias 0.0007
CE 2.453
CRPS 0.027
68% coverage 73.0%
Predicted redshift (PDF median)
DESI spec-z
DESI–Legacy: zero shot
N = 256
NMAD 0.1537
Outliers 32.4%
Bias 0.0239
CE 5.152
CRPS 0.155
68% coverage 31.2%
DESI spec-z
CLAUDS: zero shot
N = 256
NMAD 0.1345
Outliers 96.5%
Bias -0.5198
CE 19.492
CRPS 1.490
68% coverage 1.6%
ZPHOT (photometric reference)